

Handwriting Performance Before and After a Mobile Intensive Writing Clinic: A Secondary Analysis of a One-Group Kindergarten Intervention

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PUBLICATION DATE: 2022

PUBLISHER: Shirley Publishing House

ISBN: 978-621-473-043-8

ABSTRACT

This article re-examines the quantitative handwriting outcomes reported in an action research project involving 22 kindergarten pupils at Pallas Elementary School during School Year 2021-2022. The intervention used a Mobile Intensive Writing Clinic supported by writing worksheets and direct monitoring of letter formation. The source reports a pretest Mean Percentage Score (MPS) of 37.50 and a posttest MPS of 79.02, a numerical increase of 41.52 points. It also reports a computed t-value of 18.10 against a critical value of 2.08 at $\alpha = 0.05$. The present article interprets these findings cautiously because the study used a one-group pretest-posttest design without a comparison group. The evidence supports a substantial within-group improvement during the intervention period, while causal attribution remains limited by testing, maturation, history, practice, and measurement threats.

Keywords: *handwriting, kindergarten, pretest-posttest, early literacy, fine motor skills, action research, intervention*

INTRODUCTION

Handwriting is both a communication skill and a coordinated motor activity. In the source study, handwriting is described as a complex skill involving linguistic, cognitive, perceptual, and motor components. The local problem emerged during distance learning, when the researcher observed difficulties in letter recognition, letter formation and reversals, letter size and height, slant, alignment, and spacing among kindergarten pupils.

The Mobile Intensive Writing Clinic was created as a direct response to that problem. The intervention combined a mobile home-based delivery system with writing worksheets, close observation of letter strokes, and repeated practice. This article isolates the performance component of the original action research and asks what can reasonably be concluded from the reported pretest and posttest results.

Design and Participants

The original study employed a one-group pretest-posttest design. Twenty-two kindergarten pupils participated: nine boys and thirteen girls. All enrolled kindergarten pupils were included through purposive sampling. The pretest and posttest targeted the same handwriting content, including upper- and lower-case letter production and the quality of letter strokes.

The handwriting rubric considered directionality, starting position, use of writing lines, closure of circles, and straightness of lines. The study reports categories from Outstanding to Below Standard. Because the same participants were tested before and after the intervention, the design is appropriate for documenting change within

the group, but it does not by itself eliminate alternative explanations for change.

Indicator Source-reported result Interpretive note

Participants 22 kindergarten pupils Same cohort measured before and after

Pretest MPS 37.50 Source describes weak/below-standard initial performance

Posttest MPS 79.02 Source tables use inconsistent categorical labels; numerical score is retained

Numerical gain 41.52 points Large within-group increase from pretest to posttest

Computed t-value 18.10 Reported by source

Critical t-value 2.08 Reported at $\alpha = 0.05$

Source conclusion Significant difference Supports improvement during the intervention period

Interpreting the Improvement

The movement from 37.50 to 79.02 is the clearest quantitative finding. Regardless of the inconsistent descriptive category attached to 79.02 in different portions of the report, the numerical improvement is substantial. The pupils ended the intervention period performing much closer to the proficiency bands defined in the rubric than they did at baseline.

The source attributes the improvement to exposure to intensive handwriting practice. This interpretation is plausible within the action-research context because the intervention directly targeted the observed deficits: directionality, starting position, line use, circle closure, and straight-line control. It also incorporated repeated writing worksheets and direct observation of

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the child's writing process.

Methodological Caution

A one-group pretest-posttest design cannot fully separate intervention effects from other influences. Testing effects are particularly relevant because the source itself notes that shorter pretest-posttest intervals can increase testing threats. Practice, natural motor development, maturation, repeated exposure to the same target letters, home support, and teacher attention may also contribute to improvement.

For that reason, the most defensible journal statement is that handwriting performance improved significantly during the period in which the Mobile Intensive Writing Clinic was implemented. A stronger causal claim - that the clinic alone produced the full gain - would require a more rigorous comparison design or repeated-baseline approach.

Implications

The findings support continued exploration of intensive, individualized handwriting practice for early learners who demonstrate weak baseline performance. The source recommendations also call for additional formative assessments and for future work on writing readiness and visual-spatial motor skills.

A replication study could strengthen the evidence by using a larger sample, multiple schools, a comparison group, blinded scoring, repeated measures, and a clearly specified paired-samples analysis. These additions would make it possible to distinguish short-term practice effects from durable handwriting development.

CONCLUSION

The original data show a marked improvement in handwriting scores among the 22 kindergarten pupils from pretest to posttest. The source also reports a statistically significant difference. The result is promising for classroom action research, but the one-group design warrants cautious interpretation. The strongest conclusion is that handwriting performance improved during participation in the Mobile Intensive Writing Clinic, not that all improvement can be causally assigned to the intervention.

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